APPENDIX A10 · MONTE CARLO FORWARD TEST

The wheel, the next 4 and 13 weeks: 2,000 forward paths

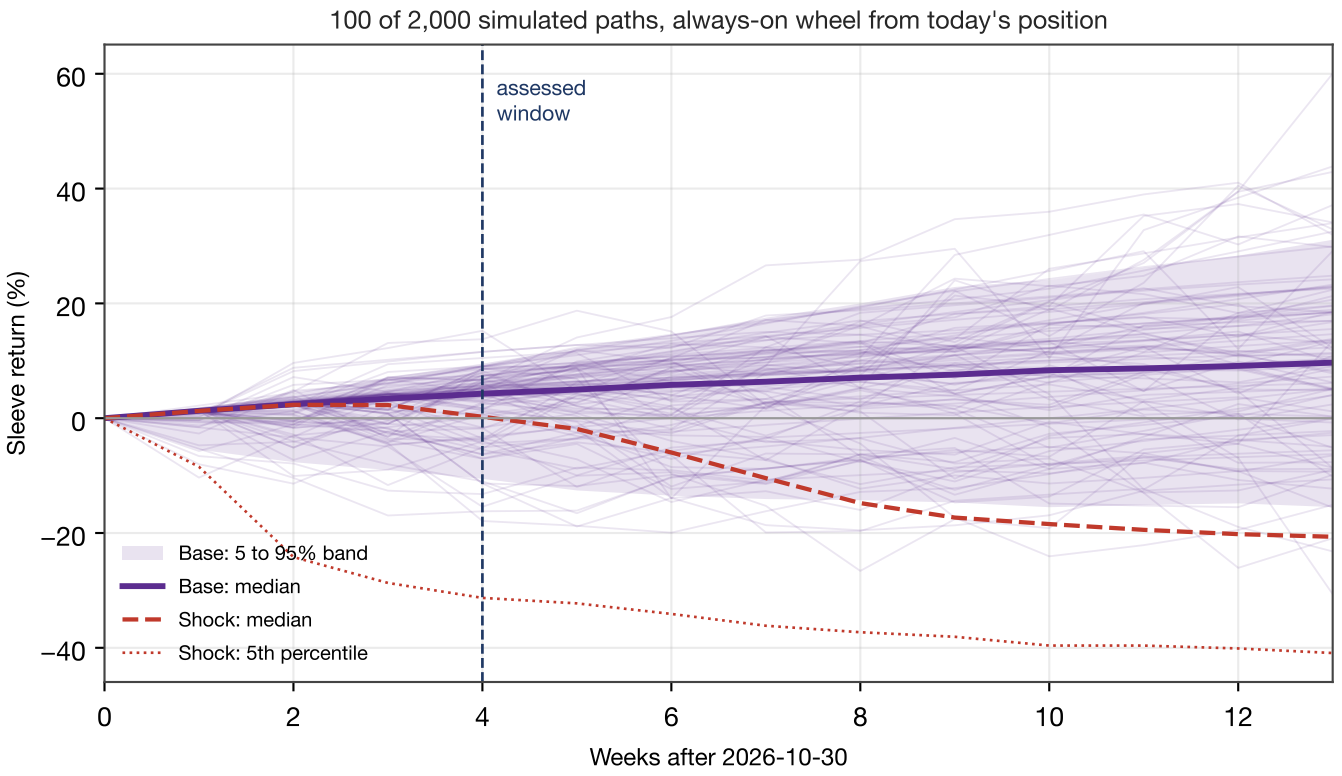
74%
Paths up after 4 weeks

+4.3%
Median 4-week return

–10.6%
5th percentile, 4 weeks

+35%/yr
Compounded, 13-week paths

80%
Past outcomes inside 5–95% band



Scenario	4wk median	4wk P(up)	13wk median	13wk 5th pct	13wk P(up)	Per year (compounded)
Base	**+4.3%**	**74%**	**+9.7%**	**–15.5%**	**74%**	**+35%**
Noise	+3.6%	70%	+6.3%	–19.3%	67%	+18%
Shock	+0.2%	50%	–20.7%	–40.9%	10%	–60%
History	+8.8%	76%	+23.9%	–21.9%	76%	+118%

Paths. Daily HYPE moves and vol changes were block-bootstrapped over all available data, December 30, 2024 through October 31, 2026, with blocks averaging 20 days. Each move is scaled by the previous day's 30-day realized vol, so realized volatility stays tied to the volatility the options are priced at. Every path starts from the current book: short a $85 put with HYPE at $89.96.

Noise. Every path draws its own option pricing, at 0.83 to 1.0 × the vol proxy, and fills at 85 to 95% of model price.

Shock. The largest weekly loss in the sample (–31%) is replayed once, at a random point within the horizon.

Calibration. Forecasting from 15 past dates, using only the data available then, put 80% of four-week outcomes inside the 5–95% band (target 90%) and 80% of thirteen-week ones. 0% of thirteen-week outcomes fell below the 5th percentile, so the thirteen-week downside is understated.

Still working, most likely: 74% of paths higher after 4 weeks, 35% a year compounded (backtest 118%). The risk is a crash week: –21% median at 13 weeks.